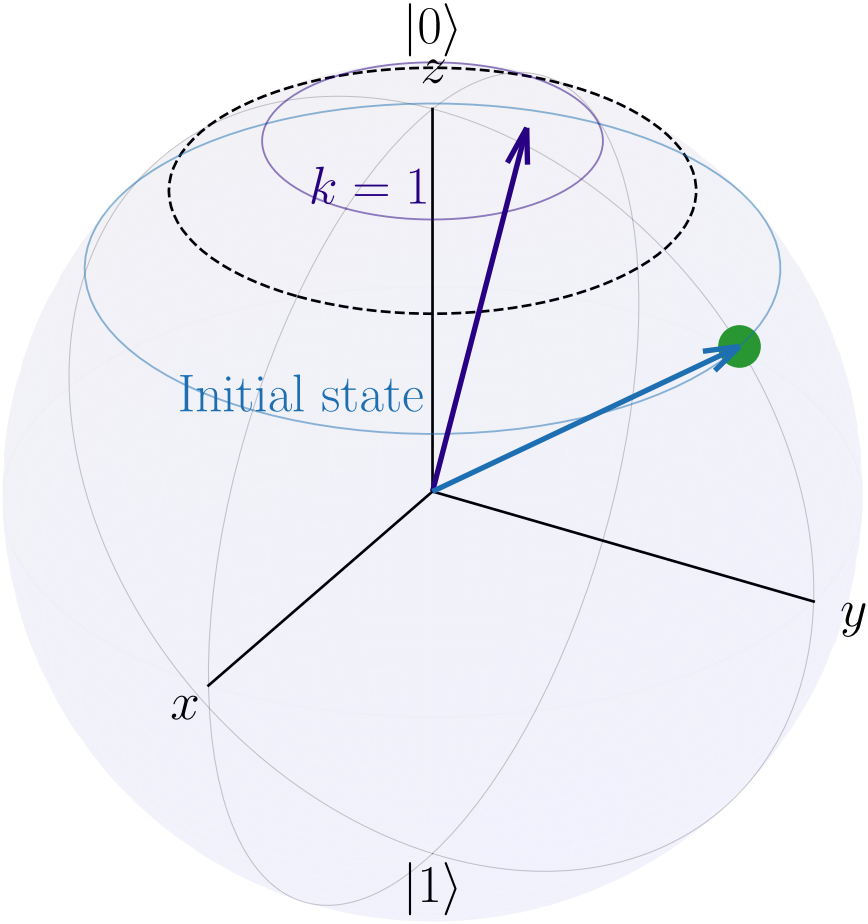

$|0\rangle$
z
$k = 1$
Initial state
y
x
$|1\rangle$